Critical-Boundary Calibration of Cosmic-Fate Probabilities

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Abstract

Posterior probabilities for cosmic-fate classes are often read as direct forecasts: one infers cosmological parameters, applies a future-fate classifier to posterior samples, and reports the resulting class fractions. This interpretation can be misleading near Lambda-CDM or any other model that lies on or close to a fate-classification boundary. We frame posterior fate probabilities as boundary statistics. In a one-dimensional calibrated boundary model, repeated experiments produce an approximately uniform distribution of posterior class probabilities, with mean 0.501, variance 0.084, and KS p = 0.165 against Uniform(0,1). A cosmology-shaped two-dimensional w0-wa analogue gives the same diagnostic behavior, with mean 0.501 and KS p = 0.058. We then define a finite-mock null-calibration protocol and apply it to an existing LCDM mock case study. In that case, the direction layer is uninformative by construction (`P_heat>0.5` in 50/100 noisy mocks), while the observed `P_RIP=0.0018` lies below 0/100 noisy LCDM mocks, giving a plus-one lower-tail p-value of 0.0099. We propose a five-layer reporting standard that separates Direction, Depth, Evidence, Boundary, and Horizon. The result is a protocol for reporting boundary-sensitive cosmic-fate probabilities, not a direct claim about the universe's final fate.

Keywords: cosmic fate; dark energy; boundary calibration; posterior class probabilities; null mocks; Lambda-CDM; w0-wa; finite-mock p-values.

1. Introduction

Cosmic-fate statements sit at an awkward interface between data analysis and long-future extrapolation. Modern cosmological data constrain distances, expansion histories, and dark-energy parameters; a fate statement then maps those inferred quantities through a classifier that asks what happens far beyond the directly observed range. It is natural to summarize such an analysis by a posterior class fraction, for example `P(RIP)`, `P(DECAY)`, `P(DS)`, or a heat-death-compatible aggregate. That number is useful, but its meaning is not the same as an ordinary posterior probability for a smooth parameter interval.

The reason is geometric. Fate labels are typically produced by threshold rules: a finite-time divergence triggers one label, an asymptotic de Sitter condition another, a collapse condition another, and an audit channel catches cases not covered by the declared rules. A posterior fate probability is therefore an integral of an indicator function over a label region. Near the boundary between two such regions, a small displacement of the posterior center can produce a large change in the reported class mass even when the underlying parameter inference is well calibrated.

This paper isolates that statistical problem. The object of study is not the real universe's final state. The object is the repeated-experiment behavior of posterior class masses when the data-generating model lies on or close to a classification boundary. This distinction matters because a boundary case can produce a direction rate near 50 percent without indicating a broken inference pipeline. Conversely, a raw class probability can still be informative if it is unusually deep in the null distribution generated by the same pipeline.

The contribution is fourfold. First, we show analytically and by simulation why a calibrated boundary posterior can produce a broad, approximately uniform distribution of class probabilities. Second, we translate the result into a cosmology-shaped w0-wa analogue and an abstract fate-classifier framework. Third, we define a finite-mock null-calibration rule that converts raw class probabilities into tail-depth statements. Fourth, we propose a reporting standard that separates Direction, Depth, Evidence, Boundary, and Horizon. This separation is the central safeguard: it prevents a boundary-sensitive posterior mass from being over-read as a direct fate forecast or as model evidence.

The manuscript uses several standard cosmological reference points only as context: CPL-like dark-energy parametrizations (Chevallier and Polarski 2001; Linder 2003), late-time data combinations such as DESI BAO and supernova samples (DESI Collaboration 2025; Brout et al. 2022), compressed CMB distance priors (Chen, Huang, and Wang 2019), and the long-standing literature on dark-energy fate questions (Dyson 1979; Krauss and Turner 1999; Krauss and Starkman 2000; Caldwell, Kamionkowski, and Weinberg 2003). The method itself does not depend on accepting any particular current cosmological claim.

2. Boundary-Statistic Formulation

Let `theta` denote the parameter vector used by a cosmological analysis. It may include `w0`, `wa`, matter density, curvature, nuisance parameters, or a richer description of the dark-energy history. Let

```
C(theta) in {CRUNCH, RIP, DS, DECAY, OTHER}
```

be a deterministic classifier that maps each parameter point to one fate label. For a label `L`, the posterior class mass is

```
q_L(data) = P(C(theta)=L | data)
          = integral 1[C(theta)=L] p(theta | data) dtheta.
```

This is a posterior probability, but it is also a statistic produced by a threshold map. Its repeated-experiment distribution can be broad when the data-generating value `theta0` lies on a boundary between labels.

For two labels `A` and `B`, write a local boundary in terms of a score

```
g(theta) = 0,
```

with `A` on one side and `B` on the other. The critical-boundary case is

```
g(theta0) = 0
```

or a near-boundary version where `g(theta0)` is small compared with the posterior uncertainty in the boundary-normal direction.

The minimal Gaussian version makes the issue explicit. Suppose

```
x_hat ~ Normal(theta0, sigma_data)
theta | x_hat ~ Normal(x_hat, sigma_post)
C(theta) = A if theta > 0, otherwise B.
```

The posterior class mass is

```
q_A = P(theta > 0 | x_hat) = Phi(x_hat / sigma_post).
```

If `theta0=0` and `sigma_post=sigma_data`, then

```
x_hat / sigma_data ~ Normal(0,1),
q_A = Phi(Z),  Z ~ Normal(0,1).
```

By the probability integral transform (Rosenblatt 1952), `q_A ~ Uniform(0,1)`. Thus a perfectly calibrated boundary null produces class probabilities spread across the full unit interval. A direction rule such as `q_A>0.5` has expected success rate 50 percent under this null. That fact is not a failure of calibration; it is the signature of a boundary statistic.

The same argument extends to a linear boundary in a multivariate Gaussian posterior. Let

```
x_hat ~ Normal(theta0, Sigma_data)
theta | x_hat ~ Normal(x_hat, Sigma_post)
g(theta) = b^T theta - c.
```

Then

```
q_A = P(g(theta)>0 | x_hat)
    = Phi((b^T x_hat - c) / sqrt(b^T Sigma_post b)).
```

If the truth lies on the boundary, `b^T theta0=c`, and the repeated-experiment variance in the boundary-normal direction matches the posterior variance, `b^T Sigma_data b = b^T Sigma_post b`, the same Uniform(0,1) result follows.

This is the mathematical core of the paper. The observable class probability is not wrong, but its interpretation changes. At a boundary, it is a calibrated random variable whose null distribution must be reported.

3. One-Dimensional Boundary Null

The first simulation implements the one-dimensional model above with `theta0=0`, `sigma_data=1`, and `sigma_post=1`. The run uses 200000 repeated experiments. The baseline diagnostics are:

Statistic	Value
Mean of `q_A`	0.500960
Variance of `q_A`	0.083559
Uniform variance `1/12`	0.083333
Direction rate `q_A>0.5`	0.502015
KS D vs Uniform(0,1)	0.002498
KS p vs Uniform(0,1)	0.164593

Figure 1 shows the corresponding histogram and empirical CDF. The central point is not merely that the mean is near one half. The important point is that the whole distribution is broad and approximately uniform. A boundary-null class probability can take values near 0.05, 0.5, or 0.95 across repeated experiments without implying that the posterior calculation itself has failed.

The same simulation suite includes off-boundary and posterior-width-mismatch checks. These are diagnostic controls. Moving the truth away from the boundary introduces direction information; making the posterior too wide or too tight distorts the class-probability distribution. Those failures are useful because they show that the uniform result is not a generic artifact. It depends on the truth being at the boundary and on the calibration of repeated-experiment and posterior uncertainty along the boundary-normal direction.

4. Cosmology-Shaped w0-wa Analogue

The second simulation translates the same idea into a two-dimensional geometry resembling the local w0-wa plane. Define coordinates centered on Lambda-CDM:

```
u = w0 + 1,
v = wa,
Lambda-CDM = (u,v) = (0,0).
```

The toy boundary score is

```
s(theta) = alpha u + v,
```

with `RIP-like` on the positive side and `DECAY-like` on the negative side. This is not a physical CPL fate classifier. It is a controlled boundary analogue in which the boundary-normal score is explicit and Lambda-CDM lies on the boundary.

For 50000 repeated experiments, the baseline diagnostics are:

Statistic	Value
Mean of `P(RIP-like)`	0.501283
Variance	0.083260
Uniform variance `1/12`	0.083333
Direction rate `P(RIP-like)>0.5`	0.505160
KS D vs Uniform(0,1)	0.005944
KS p vs Uniform(0,1)	0.058189

The same Lambda-CDM truth can yield low, near-boundary, or high posterior class masses depending on the noisy posterior center. the two-dimensional experiment selected three examples from the same repeated-experiment ensemble:

Example	w0_hat	wa_hat	`P(RIP-like)`
low RIP-like probability	-0.86259	-0.45695	0.08001
near boundary	-1.05691	0.06830	0.50000
high RIP-like probability	-1.08011	0.38820	0.92001

This result does not claim that the physical CPL fate boundary is exactly linear or that the real universe is exactly on it. It establishes the local statistical mechanism: once a posterior overlaps a classification boundary, the boundary-normal projection controls the class mass. A visually realistic w0-wa posterior can therefore yield unstable fate labels even under a stable null.

5. Fate Classifier as a Threshold Map

A physical fate classifier generally evaluates continuous future-behavior diagnostics and then returns a discrete label. We represent this by four margins:

```
crunch_margin
rip_margin
ds_margin
decay_margin
```

and a priority rule:

```
if crunch_margin > 0:
    CRUNCH
elif rip_margin > 0:
    RIP
elif ds_margin > 0:
    DS
elif decay_margin > 0:
    DECAY
else:
    OTHER
```

The labels retain their conceptual roles: `CRUNCH` for collapse, `RIP` for a finite-time divergence, `DS` for de Sitter-like asymptotics, `DECAY` for a dark-energy-decay or eternal-expansion-compatible branch, and `OTHER` as an audit channel. The method does not require the reader to accept one specific physical implementation of these labels. It requires that whatever classifier is used be treated as a threshold map with auditable boundaries.

The the classifier-framework example example posterior produces mixed class mass:

Label	Posterior fraction
CRUNCH	0.00000
RIP	0.36857
DS	0.03286
DECAY	0.50729
OTHER	0.09129

The associated path diagnostic shows a continuous descriptor path whose label jumps from `DECAY` to `RIP` as the RIP margin crosses zero. This is the classifier-level version of the boundary-statistic problem. Smooth movement in descriptors can produce discontinuous changes in labels, and posterior class masses are integrals over those discontinuous regions.

6. Finite-Mock Null Calibration

Boundary calibration requires an explicit null distribution. This is close in spirit to predictive calibration and posterior predictive checking (Dawid 1984; Rubin 1984; Gelman, Meng, and Stern 1996). For a predeclared classifier `C`, label `L`, and observed data set, compute the raw class mass

```
p_obs = P(C(theta)=L | observed data).
```

Then choose the relevant null model, generate `N` mock data sets, and run the identical inference and classification pipeline on every mock:

```
p_j = P(C(theta)=L | mock_j),  j=1,...,N.
```

The lower-tail, upper-tail, and two-sided finite-mock calibrated p-values are:

```
p_lower = (#{p_j <= p_obs} + 1) / (N + 1)
p_upper = (#{p_j >= p_obs} + 1) / (N + 1)
p_two_sided = min(1, 2 * min(p_lower, p_upper)).
```

The plus-one rule prevents zero p-values from finite mock ensembles, matching the finite randomization principle that simulation p-values should not be reported as zero (Phipson and Smyth 2010). The minimum reportable one-sided tail probability is

```
p_floor = 1 / (N + 1).
```

This calibration does not replace the raw class mass. It answers a different question. The raw mass says how much posterior probability lies in a label region. The calibrated tail depth says how unusual that raw mass is under the declared null.

The synthetic the finite-mock demonstration demonstration used 100 finite null mocks and three observed values. The results were:

Case	Raw probability	Lower-tail p	Upper-tail p	Two-sided p
Central boundary-like	0.500	0.44554	0.56436	0.89109
Low-tail example	0.018	0.03960	0.97030	0.07921
High-tail example	0.982	0.98020	0.02970	0.05941

The synthetic examples are not cosmology claims. They illustrate the reporting grammar that the real-pipeline case study then uses.

7. Real-Pipeline LCDM Mock Case Study

The case study applies the calibration protocol to existing outputs from a cosmology pipeline. It uses 100 noisy LCDM mocks as the null distribution and keeps the zero-noise mock separate as a mechanism diagnostic. The analyzed quantities are

```
P_heat = P(DS) + P(DECAY),
P_RIP  = P(RIP).
```

The noisy mocks are the finite null distribution. The zero-noise mock is not included in the null tail counts because it answers a different question: what the deterministic pipeline does when the mock data exactly match the null without noise.

The case-study results are:

Quantity	Value
Noisy LCDM mocks	100
Null `P_heat` mean	0.52515
Null `P_heat` standard error	0.03026
Null `P_heat` KS p vs Uniform	0.25266
Direction rate `P_heat > 0.5`	0.50000
Observed `P_heat`	0.99821
Observed `P_RIP`	0.001792
Noisy mocks with `P_RIP <= observed`	0 / 100
Plus-one lower-tail p for `P_RIP`	0.00990
Zero-count 95% upper bound	0.02951
Zero-noise mock000 `P_heat`	0.52275
Zero-noise mock000 `P_RIP`	0.47577
Max OTHER fraction in noisy mocks	0.001416
Max boundary fraction in noisy mocks	0.000871

The direction layer is deliberately uninformative under this LCDM boundary null: `P_heat>0.5` in 50/100 noisy mocks, and the zero-noise mock sits near a 52/48 heat/RIP split. This would look poor if judged by a high fixed direction accuracy threshold, but it is the expected boundary behavior.

The depth layer says something different. The observed `P_RIP` is lower than all 100 noisy LCDM mocks. With plus-one smoothing this becomes a one-sided lower-tail p-value of 0.00990, not zero. The zero-count 95 percent upper bound is 0.02951. This is an interpretable tail-depth statement under the declared null and finite mock count.

The correct reading is therefore narrow:

```
Under the declared LCDM null and this pipeline, the direction layer is unstable,
while the observed P_RIP is unusually low relative to the 100 noisy null mocks.
```

It is not a direct forecast of the real universe's fate. It is not, by itself, evidence that a larger dark-energy model is preferred. Those questions belong to the Evidence and Horizon layers described next.

8. Reporting Standard

The reporting unit for boundary-sensitive fate probabilities should have five layers:

Layer	Question	Required report	Failure mode
Direction	Which label has more posterior mass?	raw P(label | data); direction threshold, usually 0.5; same quantity under null mocks	Treating P > 0.5 as a discovery near a boundary.
Depth	How unusual is the raw probability under the null?	predeclared null model; mock count and finite-mock floor; lower-tail, upper-tail, and two-sided calibrated p-values	Reporting a raw tail probability without null calibration.
Evidence	Does the enlarged model deserve preference?	likelihood-ratio or delta chi2 where appropriate; AIC/BIC or Bayes factor where available; clear separation from fate-class probability	Confusing a class probability with evidence for the model.
Boundary	Is the label sensitive to thresholds or classifier defects?	boundary-flip fraction or threshold sensitivity; OTHER fraction and audit rule; known classifier limitations	Hiding that posterior mass lies near a label boundary.
Horizon	Is the fate decision inside the data-constrained region?	future scale used by the classifier; constraint horizon or equivalent diagnostic; statement when the horizon diagnostic is unavailable	Interpreting unconstrained far-future extrapolation as data.

The minimum prose should contain three distinct statements:

```
Raw class mass:
The posterior mass in label L is ...

Calibrated boundary statistic:
Under the predeclared null, the observed class probability lies in the ...
tail with finite-mock-calibrated p = ...

Guardrail:
This is not, by itself, a direct forecast of the universe's fate.
```

The the real-pipeline case study case study fills Direction, Depth, and Boundary. It does not fill Evidence and Horizon. That incompleteness is intentional. A method paper should not silently promote an available class-mass result into an unavailable model preference or extrapolation-horizon result.

For the case study, the layer fill is:

Layer	Case-study fill
Direction	Observed P_heat=0.99821; LCDM null has P_heat>0.5 in 50/100 mocks.
Depth	Observed P_RIP=0.00179; 0/100 null mocks are lower; plus-one p=0.00990, 95% CP upper=0.02951.
Evidence	Not evaluated by the real-pipeline case study figure; a science paper must add likelihood-ratio, AIC/BIC, or Bayes-factor evidence separately.
Boundary	max OTHER=0.00142; max boundary fraction=0.00087; mock000 P_heat=0.52275.
Horizon	Not evaluated by the real-pipeline case study figure; a science paper must report constraint horizon or explicitly mark it unavailable.

This is the practical contribution of the paper. The standard provides a citable format for future fate analyses: raw class mass, null-calibrated depth, model evidence, boundary sensitivity, and horizon validity must be separately visible.

9. Robustness and Diagnostic Failures

A method that only shows a favorable toy example is not enough. the robustness exercise therefore asks which perturbations preserve the boundary-null behavior and which correctly fail diagnostics.

Variant	Mean	Variance	Direction rate	KS p	Status
calibrated	0.498806	0.083597	0.497925	0.147	pass
shifted	0.499364	0.083356	0.497250	0.245	pass
off-boundary	0.638641	0.075633	0.691025	0	diagnostic failure
too wide	0.499690	0.049937	0.499387	0	diagnostic failure
too tight	0.499530	0.117285	0.498800	0	diagnostic failure
curved beta=1.5	0.524333	0.086348	0.540333	2.51e-06	curvature diagnostic

The calibrated boundary and shifted-boundary cases pass. This shows that the uniform result is not tied to the arbitrary location of the threshold; if the truth and the threshold move together and calibration is correct, the boundary-null result remains.

The off-boundary case fails, as it should. Moving the truth away from the threshold creates direction information. The posterior-width mismatch cases also fail. A posterior that is too wide becomes under-confident and has too little variance in class probability; a posterior that is too tight becomes over-confident and has too much variance. These failures are diagnostic features, not method failures.

The curved-boundary case is labeled as a curvature diagnostic. Its mean is 0.524 and its KS p-value is 2.51e-06. The method does not claim that arbitrary curvature preserves uniformity. It says that curvature, posterior calibration, and distance from the boundary must be reported or tested instead of hidden.

Finite mock resolution is another robustness constraint. For an observed raw probability around 0.018, 100 mocks have a plus-one p-value floor of 1/101 = 0.00990. The probability of seeing zero mocks below that observed value under a Uniform null is still about 0.163 at `N=100`; it falls only with larger mock ensembles. This means finite null ensembles should be reported with their resolution, not treated as unlimited tail probes.

10. Claim Ledger and Proof Boundary

The paper's claims are deliberately limited:

Status	Claim	Support or boundary
supported	Near a calibrated critical boundary, posterior class probabilities can be broad and approximately uniform under repeated experiments.	the one-dimensional experiment mean=0.50096, variance=0.08356, KS p=0.16459; the two-dimensional experiment KS p=0.05819.
supported	A 50 percent direction rate is not automatically a failed classifier near the boundary.	the real-pipeline case study LCDM mocks have P_heat>0.5 in 0.50 of mocks while the P_heat null shape passes KS p=0.25266.
supported	Tail depth can still carry information when direction is unstable.	Observed P_RIP=0.00179 is below 0/100 noisy LCDM mocks; plus-one p=0.00990.
supported	The reporting unit must separate Direction, Depth, Evidence, Boundary, and Horizon.	the reporting-standard exercise defines all five layers and marks Evidence and Horizon as unfilled in the case study.
not claimed	The method establishes the real universe's final fate.	Excluded by the problem-definition section non-goals and by the real-pipeline case study scope.
not claimed	The case study proves dynamical dark energy or model preference.	Evidence layer is explicitly unavailable in the reporting-standard exercise and must be supplied by a separate science analysis.
not claimed	Curved or miscalibrated boundaries always preserve uniformity.	the robustness exercise shows off-boundary truth and posterior-width mismatch fail; beta=1.5 curvature is labeled a diagnostic.

The negative entries are as important as the positive ones. This method paper does not establish the real universe's final fate. It does not prove dynamical dark energy or any model preference. It does not claim that curved or miscalibrated boundaries always preserve uniformity. It provides a calibration and reporting framework for a specific statistical failure mode: interpreting boundary-sensitive class masses as direct fate forecasts.

11. Limitations

The main limitation is that the toy proofs are local. The one-dimensional and linear w0-wa analogues explain the boundary-normal mechanism, but a physical fate classifier may have curved, intersecting, or prior-truncated boundaries. The response is not to discard the method; it is to report the relevant boundary diagnostics and to treat strong curvature as a diagnostic condition.

The second limitation is finite mock resolution. With 100 mocks, the smallest reportable one-sided p-value is 0.00990. This is adequate for demonstrating the reporting protocol, but not for resolving much deeper tails. Larger mock ensembles would be needed for stronger tail claims.

The third limitation is the case-study scope. The LCDM mock case demonstrates direction-depth separation for one pipeline and one declared null. It does not establish that all cosmological pipelines, data combinations, or classifiers would produce the same calibrated tail depth.

The fourth limitation is the unfilled Evidence and Horizon layers. Evidence requires likelihood-ratio, information-criterion, Bayes-factor, or equivalent model-comparison work. Horizon requires an assessment of whether the future scale used by the fate classifier lies inside a data-constrained regime. The case study intentionally does not fill those rows.

Finally, this paper does not validate compressed CMB likelihoods as equivalent to full Planck analyses, nor does it settle the current observational debate about evolving dark energy. Those are science-paper tasks. The present method paper supplies the reporting discipline needed before such results are turned into fate statements.

12. Conclusion

Posterior fate probabilities near a critical boundary should be treated as boundary statistics. In the calibrated boundary null, broad class probabilities and 50 percent direction rates are expected. That does not make the raw class mass useless; it changes the reporting question. The raw mass must be paired with a null distribution, finite-mock tail calibration, boundary diagnostics, model evidence, and an extrapolation-horizon statement.

The proposed five-layer standard gives a compact rule: report Direction, Depth, Evidence, Boundary, and Horizon separately. The LCDM mock case study shows why this matters. Direction can be unstable under the null, while an observed tail depth can still be calibrated and reported. Keeping those statements separate prevents both false pessimism about a working classifier and false confidence about the universe's final fate.

Figure Captions and Source Files

Figure	Caption target	Source
1	One-dimensional boundary-null calibration	figures/phase1_toy_boundary_uniformity.png
2	Two-dimensional w0-wa boundary analogue	figures/phase2_w0wa_boundary.png
3	Abstract fate-classifier framework	figures/phase3_classifier_framework.png
4	Finite-mock null-calibration protocol	figures/phase4_null_calibration.png
5	Real-pipeline LCDM mock case study	figures/phase5_cosmology_mock_case.png
6	Five-layer reporting standard	figures/phase6_reporting_standard.png
A1	Robustness diagnostics	figures/phase7_robustness.png

Appendix A. Robustness Details

The one-dimensional robustness checks use the same Gaussian boundary statistic as Section 3. The pass rule requires the mean to be within 0.02 of 0.5, the variance to be within 0.02 of `1/12`, the direction rate to be within 0.02 of 0.5, and the KS p-value against Uniform(0,1) to be at least 0.01. The calibrated and shifted-boundary cases pass these checks. The off-boundary, posterior-too-wide, and posterior-too-tight cases fail in the intended directions.

The curved-boundary experiment uses a two-dimensional score

```
score = v + alpha*u + beta*u^2
```

with truth at `(u,v)=(0,0)` and `beta=1.5`. This setting is intentionally strong enough to trigger a curvature diagnostic. A physical application should therefore report local boundary curvature or use mocks generated through the full classifier rather than relying only on a linear approximation.

Appendix B. Reporting Checklist

Use this checklist whenever a fate probability is reported near a critical boundary:

1. State the raw posterior class mass and the classifier used.
2. State the null model and why it is the relevant boundary or near-boundary

null.

3. State the number of mocks and the finite-mock p-value floor.
4. Report lower-tail, upper-tail, and two-sided calibrated p-values.
5. Report direction behavior under the null separately from tail depth.
6. Report model evidence separately from class mass.

7. Report boundary sensitivity, OTHER fraction, or equivalent classifier audit.
8. Report the extrapolation horizon or mark it unavailable.
9. Include the guardrail: the class mass is not by itself a direct final-fate

forecast.

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Rendered Figure Plates

Figure 1. One-dimensional boundary-null calibration

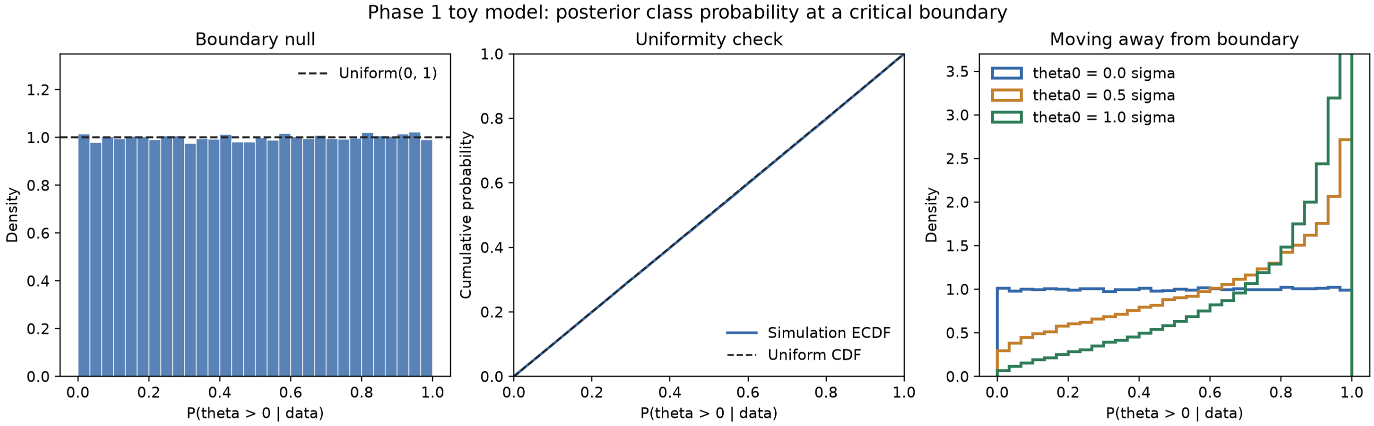

Figure 1: One-dimensional boundary-null calibration. Source file: figures/phase1_toy_boundary_uniformity.png.

Figure 2. Two-dimensional w0-wa boundary analogue

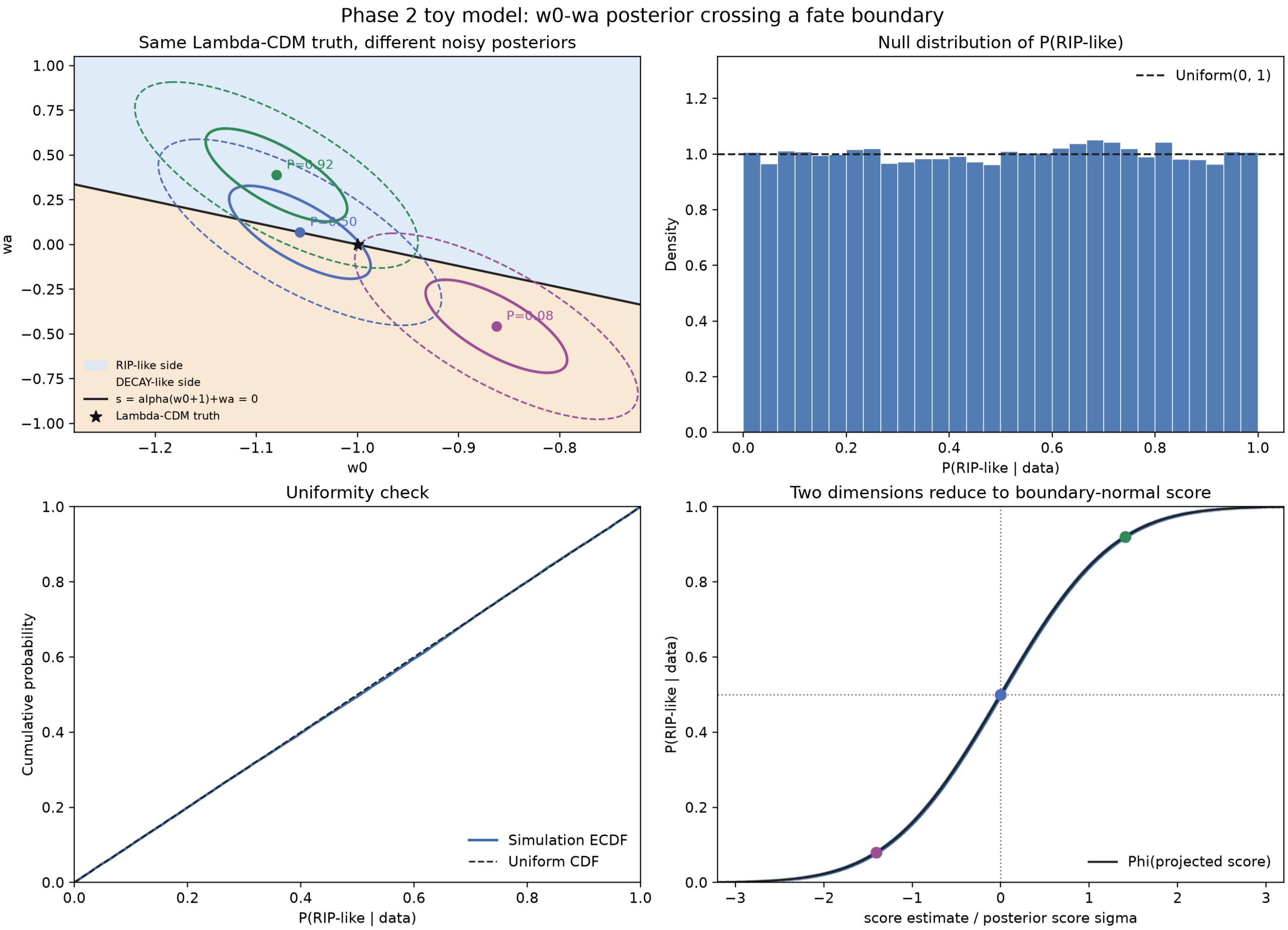

Figure 2: Two-dimensional w0-wa boundary analogue. Source file: figures/phase2_w0wa_boundary.png.

Figure 3. Abstract fate-classifier framework

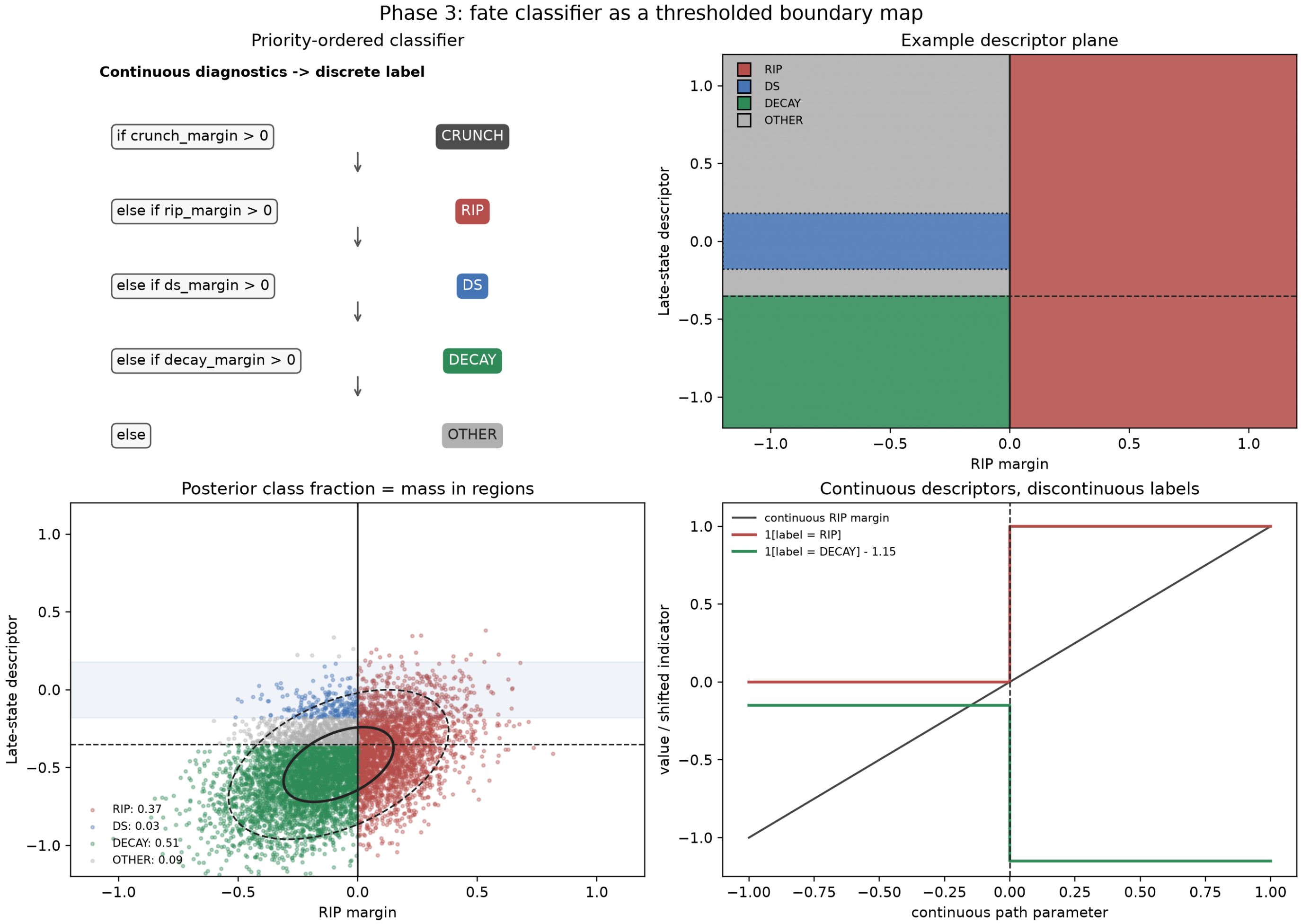

Figure 3: Abstract fate-classifier framework. Source file: figures/phase3_classifier_framework.png.

Figure 4. Finite-mock null-calibration protocol

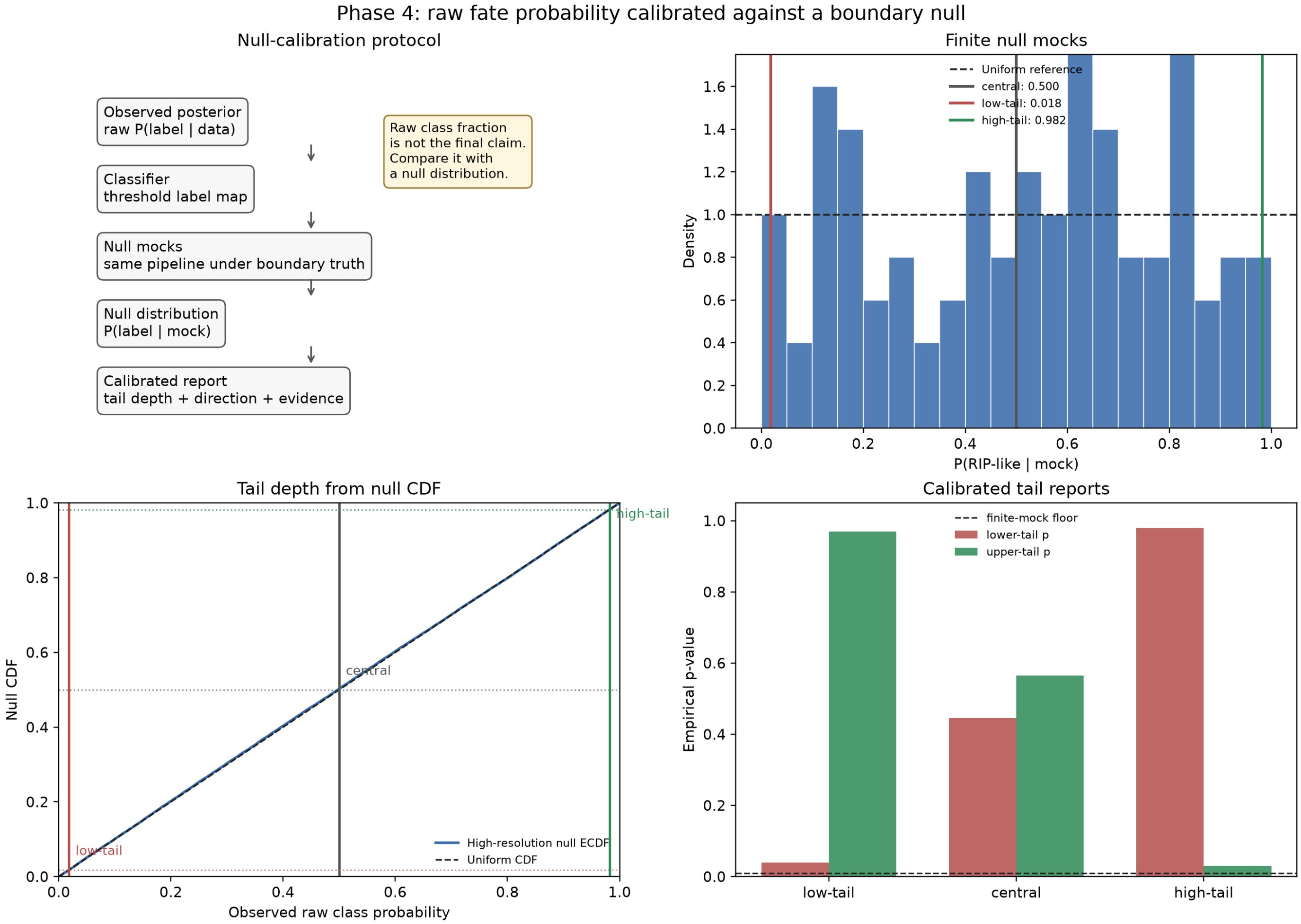

Figure 4: Finite-mock null-calibration protocol. Source file: figures/phase4_null_calibration.png.

Figure 5. Real-pipeline LCDM mock case study

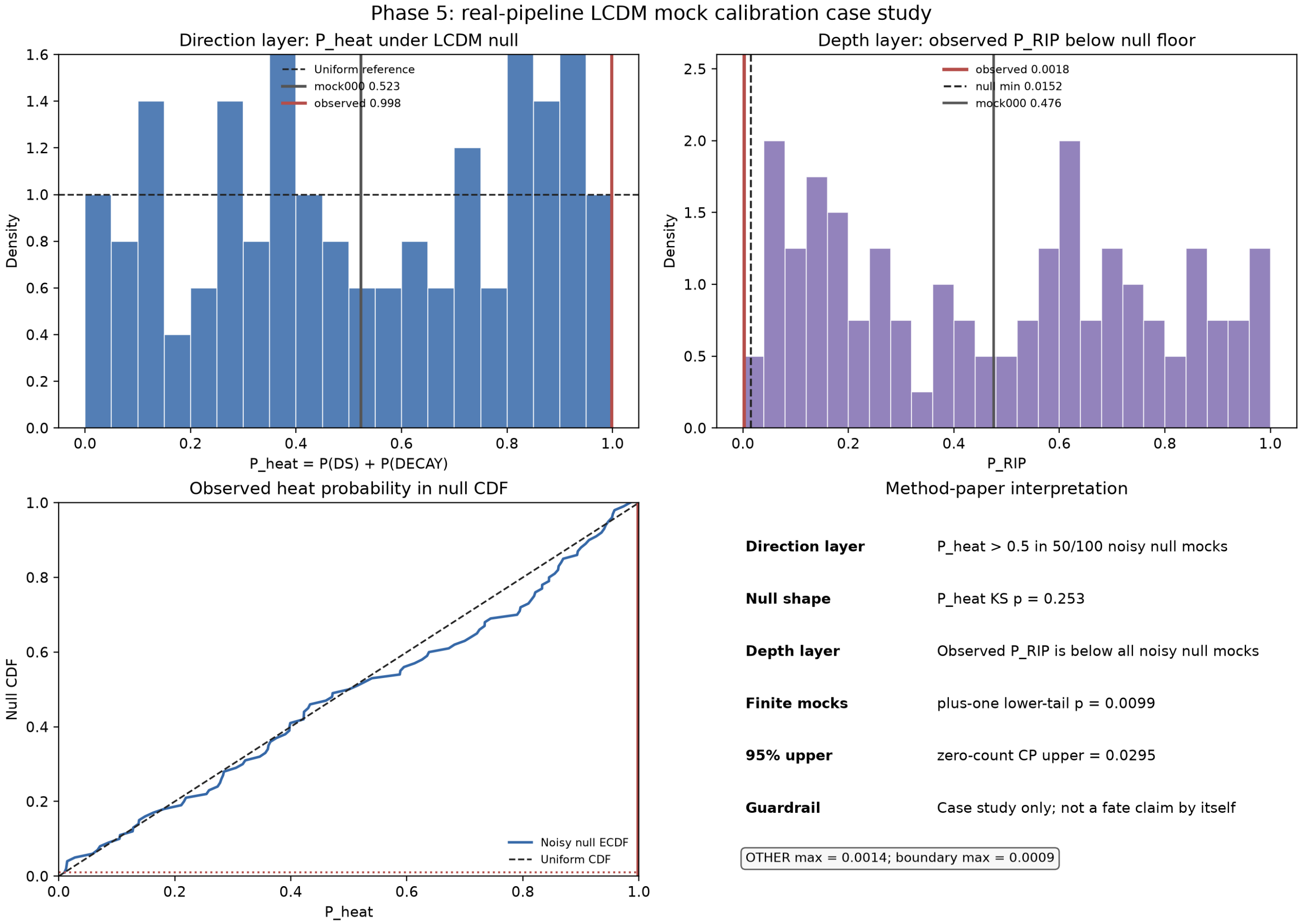

Figure 5: Real-pipeline LCDM mock case study. Source file: figures/phase5_cosmology_mock_case.png.

Figure 6. Five-layer reporting standard

Phase 6: reporting standard for calibrated fate probabilities

Layer	Question	Required report	Phase 5 case-study fill
Direction	Which label has more posterior mass?	- raw P(label | data) - direction threshold, usually 0.5 - same quantity under null mocks	Observed P_heat=0.99821; LCDM null has P_heat>0.5 in 50/100 mocks.
Depth	How unusual is the raw probability under the null?	- predeclared null model - mock count and finite-mock floor - lower-tail, upper-tail, and two-sided calibrated p-values	Observed P_RIP=0.00179; 0/100 null mocks are lower; plus-one p=0.00990, 95% CP upper=0.02951.
Evidence	Does the enlarged model deserve preference?	- likelihood-ratio or delta chi2 where appropriate - AIC/BIC or Bayes factor where available - clear separation from fate-class probability	Not evaluated by the Phase 5 case-study figure; a science paper must add likelihood-ratio, AIC/BIC, or Bayes-factor evidence separately.
Boundary	Is the label sensitive to thresholds or classifier defects?	- boundary-flip fraction or threshold sensitivity - OTHER fraction and audit rule - known classifier limitations	max OTHER=0.00142; max boundary fraction=0.00087; mock000 P_heat=0.52275.
Horizon	Is the fate decision inside the data-constrained region?	- future scale used by the classifier - constraint horizon or equivalent diagnostic - statement when the horizon diagnostic is unavailable	Not evaluated by the Phase 5 case-study figure; a science paper must report constraint horizon or explicitly mark it unavailable.

Rule: do not report a fate probability as a direct forecast until direction, depth, evidence, boundary, and horizon layers are separated.

Figure 6: Five-layer reporting standard. Source file: figures/phase6_reporting_standard.png.

Figure A1. Robustness diagnostics

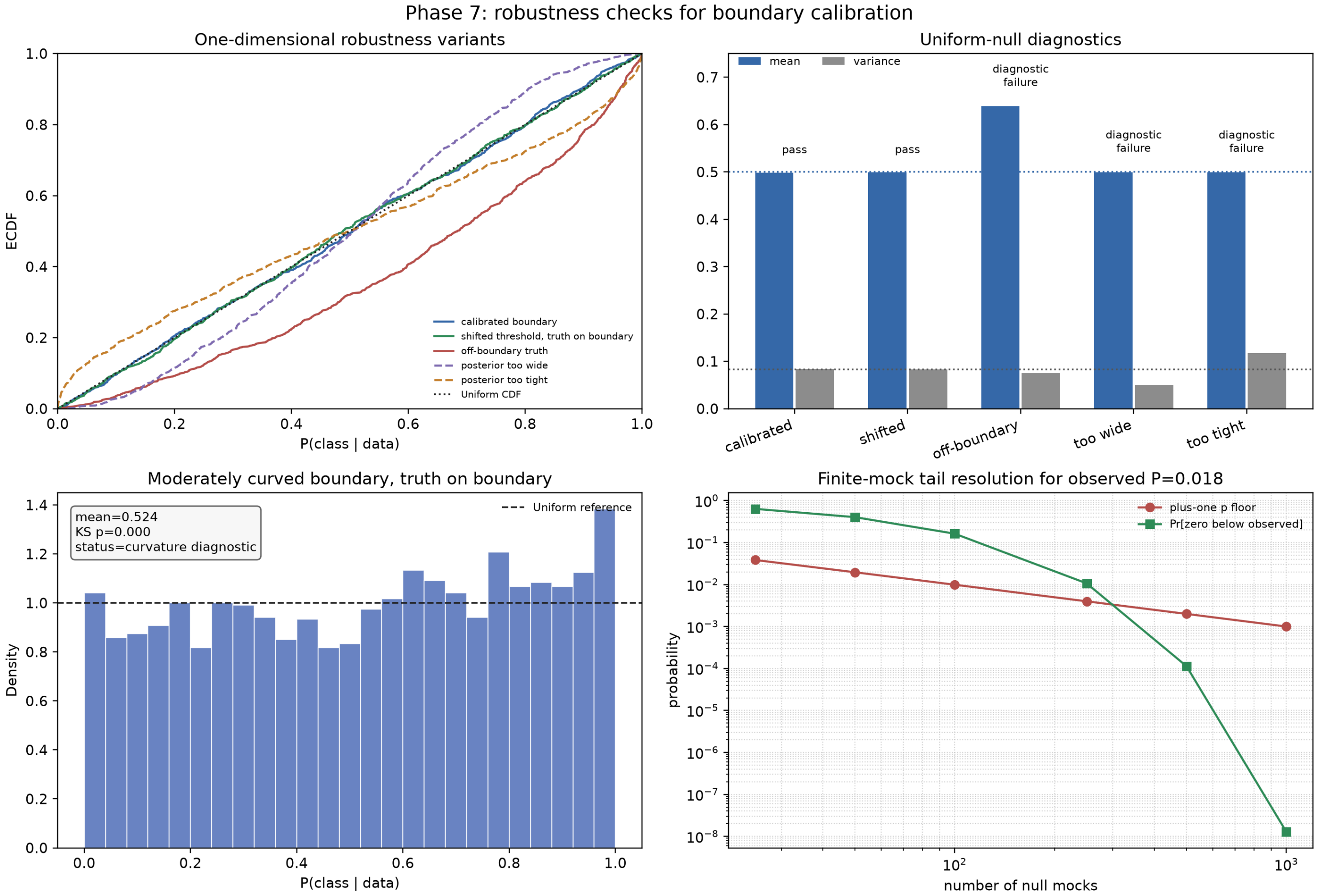

Figure A1: Robustness diagnostics. Source file: figures/phase7_robustness.png.